Дис

Sunday, 31 May 2026 12:02

Неопределени интеграли

$F(x)$ - примитивна на $f(x)$ в (a,b), ако $F'(x)=f(x)$

$\int f(x)dx$

Таблични интеграли

1) $\int x^{\alpha}dx = \frac{x^{\alpha+1}}{\alpha+1}+C$, $\alpha \neq -1$

2) $\int \frac{1}{x}dx = \ln|x|+C$

3) $\int a^x dx = \frac{a^x}{\ln a}+C$, $a>0$

3.1) $\int e^x dx = e^x + C$

4) $\int \sin x\, dx = \cos x + C$

5) $\int \cos x\, dx = \sin x + C$

6) $\int \frac{1}{1+x^2}dx = \operatorname{arctg} x + C$

7) $\int \frac{1}{\sqrt{1-x^2}}dx = \arcsin x + C$

8) $\int \frac{1}{\cos^2 x}dx = \operatorname{tg} x + C$

9) $\int \frac{1}{\sin^2 x}dx = -\operatorname{cotg} x + C$

10) $\int \frac{1}{\sqrt{x^2+a}}dx = \ln|x+\sqrt{x^2+a}|+C$

Св-ва

1) $\int c\, f(x)dx = c\int f(x)dx$

2) $\int f(x)\pm g(x)dx = \int f(x)dx + \int g(x)dx$

3) внасяне на ф-ия под диференциала

$\int g(f(x))\cdot f'(x)dx = \int g(f(x))\,df(x) \overset{\text{полагаме } f(x)=y}{=} \int g(y)dy$

$f'(x)dx$ (диференц.)

4) интегриране по части: $\int f dg = fg - \int g df$

$\int f g' dx = f\cdot g - \int g f' dx$

(*) $\int P(x)\cdot \begin{Bmatrix} e^x \\ \sin x \\ \cos x \end{Bmatrix} dx =$ (по части)

$\int \begin{Bmatrix} \ln x \\ \operatorname{arctg} x \\ \arcsin x \end{Bmatrix} P(x)\,dx =$ (по части)

не можем лесно да интегр., но можем лесно да диф.

5) Смяна на променливите

$\int f(x)dx = \int f(\varphi(t))\,d\varphi(t) = \int f(\varphi(t))\,\varphi'(t)dt$

$x=\varphi(t)$ - обратима и $\exists\, \varphi'(t)$

6) интегриране на рационални ф-ии

$\int \frac{P(x)}{Q(x)}dx$ $P(x), Q(x)$ - полиноми

- отделяне на цяла част $\int \frac{P(x)}{Q(x)}dx = \int A(x) + \frac{P_1(x)}{Q(x)}dx$ $\deg(P)<\deg(Q)$
- $\frac{P_1(x)}{Q(x)}$ → сума на елем. дроби

$Q(x) = \underbrace{(x-a)^{k_1}(x-b)^{k_2}\dots}_{\text{реални корени}}\ \underbrace{(x^2+px+q)(x^2+5x+t)^{e_2}\dots}_{\text{неразложими}}$

$\frac{A_1}{x-a}, \frac{A_2}{(x-a)^2}, \dots, \frac{A_{k_1}}{(x-a)^{k_1}}$ $\qquad \frac{M_1x+N_1}{x^2+px+q}, \dots, \frac{M_{e}x+N_{e}}{(x^2+px+q)^{e}}$

7) Интегриране на ирационални ф-ии:

$\int R(x^{\frac{p_1}{q_1}}, x^{\frac{p_2}{q_2}}, \dots, x^{\frac{p_n}{q_n}})dx =$ (рационална ф-ия)

рационална ф-ия на корени на x

$\boxed{x=t^k,\ k=\text{НОК}(q_1,q_2,\dots,q_n)}$

①
$\int x^2 + \frac{2}{1+x^2} - 3\cos x\, dx =$

$= \int x^2dx + 2\int \frac{1}{1+x^2}dx - 3\int \cos x\, dx =$

$= \frac{x^3}{3} + 2\operatorname{arctg} x - 3\sin x + C$

②
$\int \frac{x^4-\sqrt{x}+3x}{x^2}dx = \int \frac{x^4}{x^2} - \frac{\sqrt{x}}{x^2} + \frac{3x}{x^2}dx =$

$= \int x^2dx - \int x^{-\frac{3}{2}}dx + 3\int \frac{1}{x}dx =$

$= \frac{x^3}{3} - \frac{x^{-\frac{1}{2}}}{-\frac{1}{2}} + 3\ln|x| + C =$

$= \frac{x^3}{3} + \frac{2}{\sqrt{x}} + 3\ln|x| + C$

③ $(x^2)'=2x,\ 2xdx=dx^2$

$\int 2x e^{x^2}dx = \int e^{x^2}dx^2 \overset{y=x^2}{=} \int e^y dy = e^y + C \overset{y=x^2}{=} e^{x^2}+C$

④
$\int \frac{x}{x^2+1}dx = \frac{1}{2}\int \frac{1}{x^2+1}d(x^2+1) = \frac{1}{2}\ln(x^2+1)+C$

произв. на конст. е 0 => никаква промяна

$x\,dx = d\frac{x^2}{2} = \frac{1}{2}dx^2 = \frac{1}{2}d(x^2+1)$

⑤
$\int \frac{1}{4+x^2}dx = \int \frac{1}{4(1+\frac{x^2}{4})}dx = \int \frac{1}{4}\cdot\frac{1}{1+(\frac{x}{2})^2}dx = \frac{1}{2}\int \frac{1}{1+(\frac{x}{2})^2}d(\frac{x}{2}) =$

$\boxed{\int \frac{1}{1+y^2}dy = \operatorname{arctg} y + C}$ $\qquad = \frac{1}{2}\operatorname{arctg}(\frac{x}{2})+C$

⑥
$\int \frac{1}{\sqrt{3-x^2}}dx = \int \frac{1}{\sqrt{3}\sqrt{1-\frac{x^2}{3}}}dx = \int \frac{1}{\sqrt{1-(\frac{x}{\sqrt{3}})^2}}d(\frac{x}{\sqrt{3}}) =$

$= \arcsin(\frac{x}{\sqrt{3}})+C$

⑦
$\int \frac{e^{\frac{1}{x}}}{x^2}dx = \int \frac{1}{x^2}e^{\frac{1}{x}}dx = -\int e^{\frac{1}{x}}d(\frac{1}{x}) = -(e^{\frac{1}{x}}+C) =$

$\boxed{\frac{1}{x^2}dx = d(\frac{x^{-1}}{-1}) = -d(\frac{1}{x})}$ $\qquad = -e^{\frac{1}{x}}+C$ защо?

⑧
$\int x^2\cos(x^3)dx = \int \cos(x^3)\,d\frac{x^3}{3} = \frac{1}{3}\int \cos(x^3)dx^3 =$

$= \frac{1}{3}\sin(x^3)+C$

⑨ ако го виждаме на теста

$\int \cos^2 x\, dx = \int \frac{1+\cos 2x}{2}dx = \int \frac{1}{2}dx + \int \frac{\cos 2x}{2}dx\frac{2}{2} =$

$= \frac{1}{2}x + C + \frac{1}{4}\int \cos 2x\, d(2x) = \frac{1}{2}x + \frac{1}{4}\sin 2x + C$

⑩
$\int \underbrace{\operatorname{arctg} x}_{f}\cdot\underbrace{1dx}_{g} = x\cdot\operatorname{arctg} x - \int x\, d\operatorname{arctg} x =$

$= x\cdot\operatorname{arctg} x - \int x\frac{1}{1+x^2}dx = x\operatorname{arctg} x - \frac{1}{2}\int \frac{1}{1+x^2}d(x^2+1) =$

$= x\operatorname{arctg} x - \frac{1}{2}\ln(x^2+1)+C$ $\qquad \boxed{(f(g(x)))' = f'(g(x))\cdot g'(x)}$

⑪
$\int x\ln(3x)dx = \int \underbrace{\ln(3x)}_{f}\,\underbrace{d\frac{x^2}{2}}_{g} = \ln(3x)\cdot\frac{x^2}{2} - \int \frac{x^2}{2}d(\ln(3x)) =$

$= \ln(3x)\cdot\frac{x^2}{2} - \frac{1}{2}\int x^2\cdot\frac{1}{3x}\cdot 3\,dx = \ln(3x)\cdot\frac{x^2}{2} - \frac{1}{2}\int x\,dx =$

произв. на 3x, произв. на ln

$= \ln(3x)\frac{x^2}{2} - \frac{1}{2}\frac{x^2}{2}+C$

⑫
$\int x\sin x\, dx = \int x\, d(-\cos x) = -x\cos x + \int \cos x\, dx =$

$= -x\cos x + \sin x + C$

⑬
$\int e^{2x}x\, dx = \int \frac{e^{2x}}{2}\cdot 2x\, dx = \frac{1}{2}\int e^{2x}x\, d2x = \frac{1}{2}\int x\, de^{2x} =$

$= \frac{1}{2}xe^{2x} - \frac{1}{2}\int e^{2x}dx\frac{2}{2} = \frac{1}{2}xe^{2x} - \frac{1}{4}\int e^{2x}d2x =$

$= \frac{1}{2}xe^{2x} - \frac{1}{4}e^{2x}+C$

⑭
$I = \int e^x\sin x\, dx = \int \sin x\, de^x = e^x\sin x - \int e^x d\sin x =$

кое ще избера

$= e^x\sin x - \int e^x\cos x\, dx = e^x\sin x - \int \cos x\, de^x =$

избираме e^x да го внесем обратно

$= e^x\sin x - \cos x\, e^x + \int e^x d\cos x = e^x\sin x - e^x\cos x + \int e^x(-\sin x)dx$

$I = e^x\sin x - e^x\cos x - I$

$2I = e^x\sin x - e^x\cos x$

$I = \frac{e^x\sin x - e^x\cos x}{2} + C$

⑮
$\int \frac{e^{\frac{1}{x}}}{x^2}dx = \int \frac{e^t}{\frac{1}{t^2}}d(\frac{1}{t}) = \int \frac{e^t}{\frac{1}{t^2}}\cdot(-\frac{1}{t^2})dt = -\int e^t dt = -e^t + C$

$\boxed{t=\frac{1}{x} \Leftrightarrow x=\frac{1}{t}}$ $\qquad = -e^{\frac{1}{x}}+C$

⑯
$\int \frac{1}{1+e^x}dx = \int \frac{1}{1+t}\cdot\frac{1}{t}dt = \int \frac{1}{t} - \frac{1}{1+t}dt = \int \frac{1}{t}dt - \int \frac{1}{1+t}d(t+1) =$

$\boxed{t=e^x \Leftrightarrow x=\ln t \\ dx = d\ln t = \frac{1}{t}dt}$ $\qquad = \ln|t| - \ln(t+1) + C = \ln e^x - \ln(e^x+1) + C =$

$= x - \ln(1+e^x)+C$

⑰
$\int \frac{\operatorname{arctg}\sqrt{x}}{\sqrt{x}(1+x)}dx = \int \frac{\operatorname{arctg} t}{t(1+t^2)}2t\,dt = 2\int \frac{\operatorname{arctg} t}{1+t^2}dt$

$\boxed{t=\sqrt{x} \Leftrightarrow x=t^2 \\ dx = 2t\,dt}$ $\qquad = 2\int \operatorname{arctg} t\, d\operatorname{arctg} t \overset{y=\operatorname{arctg} t}{=} 2\int y\, dy =$

$= \cancel{2}\frac{y^2}{\cancel{2}} + C = (\operatorname{arctg} t)^2 = (\operatorname{arctg}\sqrt{x})^2$

⑱ Субституция на Хорнер

$\int \frac{1}{x^2+2x+2}dx = \int \frac{1}{(x+1)^2+1}dx = \int \frac{1}{t^2+1}dt = \operatorname{arctg} t + C = \operatorname{arctg}(x+1)+C$

$\boxed{t=x+1 \\ dt=dx}$

$\int \frac{1}{y^2+1}dy$

нещо първа степен искаме да елиминираме и вдигаме => доп. до пълен кв.

$x^2+2x+2 = x^2+2x\cdot 1+1^2+1 =$

$= (x+1)^2+1$

⑲
$\int \frac{5x^3+3x^2-22x-8}{x^3-4x}dx = \int 5 + \frac{3x^2-2x-8}{x^3-4x}dx$

$5x^3+3x^2-22x-8 \,|\, x^3-4x = 5$

$-5x^3 \qquad -20x$

$3x^2-2x-8$

$2<3$ => прекратяваме

$Q(x) = x^3-4x = x^1(x-2)^1(x+2)^1$ за всеки множ. ще имаме една съответна елементарна дроб

$\frac{3x^2-2x-8}{x(x-2)(x+2)} = \frac{A}{x} + \frac{B}{x-2} + \frac{C}{x+2}$ } общ знам.

$3x^2-2x-8 = A(x-2)(x+2) + Bx(x+2) + Cx(x-2)$

I начин: заместваме корените на x

$x=0: \ -8=-4A \Rightarrow \boxed{A=2}$

$x=2: \ 24=8B \Rightarrow \boxed{B=3}$

$x=-2: \ 32=8C \Rightarrow \boxed{C=4}$

II начин: $3x^2-2x-8 = A(x^2-4) + B(x^2+2x) + C(x^2-2x)$

$x^2: \ 3=A+B+C$

$x^1: \ -2=2B-2C$

$x^0: \ -8=-4A$

} решаваме линейната с-ма

Връщаме и към интеграла:

заместваме с конст., които сме намерили

$\int 5 + \frac{2}{x} + \frac{3}{x-2} + \frac{4}{x+2}dx = 5x + 2\ln|x| + 3\ln|x-2| + 4\ln|x+2| + C$

⑳ 2019 изпит

$\int \frac{x-2}{x(x^2+2)}dx = \int -\frac{1}{x} + \frac{1\cdot x+1}{x^2+2}dx = \int -\frac{1}{x}dx + \frac{1}{2}\int \frac{2x}{x^2+2}dx + \int \frac{1}{x^2+2}dx =$

$\frac{x-2}{x(x^2+2)} = \frac{A}{x} + \frac{Bx+C}{x^2+2}$

$x-2 = A(x^2+2) + Bx^2 + Cx$

II начин: $x^2: \ 0=A+B \Rightarrow 0=B-1 \Rightarrow \boxed{B=1}$

$x^1: \ 1=C \Rightarrow \boxed{C=1}$

$x^0: \ -2=2A \Rightarrow \boxed{A=-1}$

$= -\ln|x| + \frac{1}{2}\int \frac{1}{x^2+2}d(x^2+2) + \int \frac{1\cdot\sqrt{2}}{2((\frac{x}{\sqrt{2}})^2+1)}d\frac{x}{\sqrt{2}} =$

$= -\ln|x| + \frac{1}{2}\ln(x^2+2) + \frac{\sqrt{2}}{2}\cdot\operatorname{arctg}\frac{x}{\sqrt{2}} + C$

㉑
$\int \frac{x^3-2x+2}{(x-1)^2(x^2+1)}dx$

$\frac{x^3-2x+2}{(x-1)^2(x^2+1)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{Cx+D}{x^2+1}$

$x^3-2x+2 = A(x-1)(x^2+1) + B(x^2+1) + (Cx+D)(x-1)^2$

$x^3-2x+2 = A(x^3+x-x^2-1) + B(x^2+1) + (Cx+D)(x^2-2x+1)$

$x^3-2x+2 = Ax^3 - Ax^2 + Ax - A + Bx^2 + B + Cx^3 - 2Cx^2 + Cx + Dx^2 - 2Dx + D$

$x=1: \ 1=2B \Rightarrow B=\frac{1}{2}$

$x=0: \ 2=-A+B+D = -A-D+\frac{1}{2} \Rightarrow A-D=-\frac{3}{2}$

$x=2: \ 6=5A+5B+2C+D \Rightarrow 6=5A+5\frac{1}{2}+2C+A+\frac{3}{2}$

$\Rightarrow 6=6A+2C+4$

$\Rightarrow 3A+C=1 \Rightarrow -C=3A-1$

$x=-1: \ 3=-4A+2B-4C+4D$

$3=-\cancel{4A}+\cancel{2\frac{1}{2}}-12A-4+\cancel{4A}+4\frac{3}{2}$

$12A=0 \Rightarrow A=0$

$\Rightarrow C=1$

$\Rightarrow D=\frac{3}{2}$

Връщаме в интеграла:

$= \int \frac{0}{x-1} + \frac{\frac{1}{2}}{(x-1)^2} + \frac{1\cdot x+\frac{3}{2}}{x^2+1}dx = \frac{1}{2}\int \frac{1}{(x-1)^2}dx + \frac{1}{2}\int \frac{2x}{x^2+1}dx + \frac{3}{2}\int \frac{1}{x^2+1}dx =$

$= \frac{1}{2}\frac{(x-1)^{-1}}{-1} + \frac{1}{2}\ln(x^2+1) + \frac{3}{2}\operatorname{arctg} x + C$

㉒
$\int \frac{\sqrt{x}-1}{3x+\sqrt[3]{x^2}}dx = \int \frac{t^3-1}{3t^6+t^{6(\frac{2}{3})}}\cdot 6t^5dt = 6\int \frac{(t^3-1)t^5}{3t^6+t^4}dt = 6\int \frac{t^4-t}{3t^2+1}dt$

$\boxed{\sqrt{x}=x^{\frac{1}{2}} \\ \sqrt[3]{x^2}=x^{\frac{2}{3}} \\ x=t^6 \to dx=dt^6=6t^5dt \\ t=\sqrt[6]{x}}$ НОК$(2,3)=6$

Отделяме цяла част:

$t^4 \quad -t \,|\, 3t^2+1 = \frac{1}{3}t^2 - \frac{1}{9}$

$-t^4 + \frac{1}{3}t^2$

$-\frac{1}{3}t^2 - t$

$-\frac{1}{3}t^2 \quad -\frac{1}{9}$

$-t + \frac{1}{9}$

В интеграла:

$= 6\int \frac{1}{3}t^2 - \frac{1}{9} + \frac{-t+\frac{1}{9}}{3t^2+1}dt = 6\int \frac{1}{3}t^2dt - 6\int \frac{1}{9}dt - 6\int \frac{t}{3t^2+1}dt + 6\cdot\frac{1}{9}\int \frac{1}{3t^2+1}dt =$

$= 2\frac{t^3}{3} - \frac{2}{3}\cdot t - \int \frac{1}{3t^2+1}d(3t^2+1) + \frac{2}{3}\int \frac{1}{(\sqrt{3}t)^2+1}dt\frac{\sqrt{3}}{\sqrt{3}} =$

$= \frac{2t^3}{3} - \frac{2}{3}t - \ln(3t^2+1) + \frac{2}{3\sqrt{3}}\int \frac{1}{(\sqrt{3}t)^2+1}d\sqrt{3}t = \frac{2}{3}t^3 - \frac{2}{3}t - \ln(3t^2+1) + \frac{2}{3\sqrt{3}}\operatorname{arctg}(\sqrt{3}t) + C =$

$= \frac{2}{3}\sqrt{x} - \frac{2}{3}\sqrt[6]{x} - \ln(3\sqrt[3]{x}+1) + \frac{2}{3\sqrt{3}}\operatorname{arctg}(\sqrt{3}\sqrt[6]{x}) + C$

㉓ 2021 $(x>0)$

$\int \underbrace{\operatorname{arctg}\sqrt{x}}_{f}\underbrace{dx}_{g} = x\operatorname{arctg}\sqrt{x} - \int x\, d\operatorname{arctg}\sqrt{x} = x\operatorname{arctg}\sqrt{x} - \int x\cdot\underbrace{\frac{1}{1+(\sqrt{x})^2}}_{\operatorname{arctg}}\cdot\underbrace{\frac{1}{2}\frac{1}{\sqrt{x}}}_{\sqrt{x}}dx =$

$= x\operatorname{arctg}\sqrt{x} - \underbrace{\int \frac{\sqrt{x}}{2(1+x)}dx}_{I} = x\operatorname{arctg}\sqrt{x} - \sqrt{x} + \operatorname{arctg}\sqrt{x} + C$

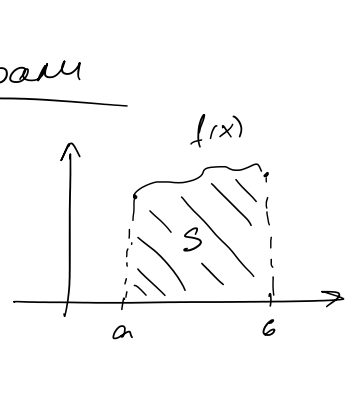
$\boxed{t=\sqrt{x} \\ x=t^2 \\ dx=2t\,dt}$

$I = \int \frac{t}{\cancel{2}(1+t^2)}\cancel{2}t\,dt = \int \frac{t^2+1-1}{(1+t^2)}dt = \int 1 - \frac{1}{t^2+1}dt = t - \operatorname{arctg} t + C =$

$= \sqrt{x} - \operatorname{arctg}\sqrt{x} - C$

Определени интеграли

$\int_a^b f(x)dx$



Ф-ла на Лайбниц - Нютон

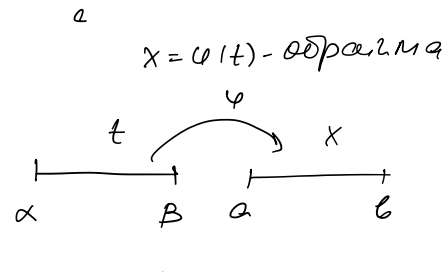
$f(x)$ е непрекъсната в $[a,b]$ и $\Phi(x)$ е нейната примитивна

$\int_a^b f(x)dx = \underbrace{\Phi(x)}_{\text{неопр. инт.}}\Big|_a^b = \Phi(b) - \Phi(a)$

по части: $\int_a^b f dg = fg\Big|_a^b - \int_a^b g df$

смяна на пром.: $\int_a^b f(x)dx$

$x=\varphi(t)$ - обратима и $\exists\varphi'(t)$



$a=\varphi(\alpha)$

$b=\varphi(\beta)$

$\Rightarrow \int_a^b f(x)dx = \int_\alpha^\beta f(\varphi(t))\varphi'(t)dt$

㉔
$\int_0^1 \frac{3}{1+x^2} + \sin x - x^4 dx =$

$= \left(3\operatorname{arctg} x + (-\cos x) - \frac{x^5}{5}\right)\Big|_0^1 =$

$= \left(3\operatorname{arctg} 1 - \cos 1 - \frac{1}{5}\right) - \left(3\operatorname{arctg} 0 - \cos 0 - 0\right) =$

$= \frac{3\pi}{4} - \cos 1 - \frac{1}{5} + 1.$

㉔ 2018
$\int_0^1 x^2\operatorname{arctg} x\, dx = \int_0^1 \underbrace{\operatorname{arctg} x}_{f}\, d\underbrace{\frac{x^3}{3}}_{g} = \frac{x^3}{3}\operatorname{arctg} x\Big|_0^1 - \int_0^1 \frac{x^3}{3}d\operatorname{arctg} x =$

$= \frac{1}{3}\operatorname{arctg} 1 - 0\operatorname{arctg} 0 - \int_0^1 \frac{x^3}{3}\frac{1}{1+x^2}dx =$

$= \frac{1}{3}\frac{\pi}{4} - 0 - \frac{1}{3}\int_0^1 \frac{x^3+x-x}{1+x^2}dx = \frac{\pi}{12} - \frac{1}{3}\int_0^1 \frac{x(x^2+1)-x}{x^2+1}dx =$

$= \frac{\pi}{12} - \frac{1}{3}\int_0^1 x - \frac{x}{x^2+1}dx = \frac{\pi}{12} - \frac{1}{3}\int_0^1 x\,dx + \frac{1}{3}\frac{1}{2}\int_0^1 \frac{2x}{x^2+1}dx =$

$= \frac{\pi}{12} - \frac{1}{3}\left(\frac{x^2}{2}\right)\Big|_0^1 + \frac{1}{6}\int_0^1 \frac{1}{x^2+1}d(x^2+1) = \frac{\pi}{12} - \left(\frac{1}{6} - 0\right) + \frac{1}{6}\ln(x^2+1)\Big|_0^1 =$

$= \frac{\pi}{12} - \frac{1}{6} + \frac{1}{6}(\ln 2 - \overset{0}{\ln 1}) = \frac{\pi}{12} - \frac{1}{6} + \frac{1}{6}\ln 2$

㉕ 2019 когато един единствен корен, полагаме него.

$\int_1^8 \frac{dx}{\sqrt[3]{x}+x} = \int_1^2 \frac{1}{t+t^3}3t^2dt = \frac{1}{2}\int_1^2 \frac{3t\cdot 2}{1+t^2}dt = \frac{3}{2}\int_1^2 \frac{1}{1+t^2}d(t^2+1) =$

$\boxed{t=\sqrt[3]{x} \\ x=t^3 \\ dx=3t^2dt \\ x=1 \Rightarrow t=\sqrt[3]{1}=1 \\ x=8 \Rightarrow t=\sqrt[3]{8}=2}$

като имаме опр. интеграл нямa нужда да и връщаме назад

$= \frac{3}{2}\ln(1+t^2)\Big|_1^2 =$

$= \frac{3}{2}\ln 5 - \frac{3}{2}\ln 2 =$

$= \frac{3}{2}\ln\frac{5}{2}$

㉖ 2017
$\int_1^2 \frac{x+1}{\sqrt{x}(1+\sqrt{x})}dx = \int_1^{\sqrt{2}} \frac{t^2+1}{t(1+t)}2t\,dt = 2\int_1^{\sqrt{2}} \frac{t^2+1}{1+t}dt = 2\int_1^{\sqrt{2}} t - 1 + \frac{2}{1+t}dt =$

$\boxed{t=\sqrt{x} \\ x=t^2 \\ dx=2t\,dt \\ x=1 \Rightarrow t=\sqrt{1}=1 \\ x=2 \Rightarrow t=\sqrt{2}}$

$t^2 \quad +1 \,|\, 1+t = t-1$

$-t^2+t$

$-t+1$

$-\ -t-1$

②

$= 2\left(\frac{t^2}{2} - t + 2\ln(1+t)\right)\Big|_1^{\sqrt{2}} = 2\left(\frac{\cancel{2}}{\cancel{2}} - \sqrt{2} + 2\ln(1+\sqrt{2})\right) - 2\left(\frac{1}{2} - 1 + 2\ln(1+1)\right) =$

$= 2 - 2\sqrt{3} + 4\ln(1+\sqrt{2}) + 1 - 4\ln 2 = 3 - 2\sqrt{2} + 4\ln\left(\frac{1+\sqrt{2}}{2}\right)$

㉗ 2019
$\int_0^2 \ln(x^2+4)dx = x\ln(x^2+4)\Big|_0^2 - \int_0^2 x\, d\ln(x^2+4) =$

$= 2\ln 8 - 0 - \int_0^2 \frac{x}{x^2+4}2x\,dx = 2\ln 8 - 2\int_0^2 \frac{x^2+4-4}{x^2+4}dx =$

$= 2\ln 8 - 2\int_0^2 \frac{x^2+4}{x^2+4} - \frac{4}{x^2+4}dx = 2\ln 8 - 2\int_0^2 1dx + 8\int_0^2 \frac{1}{x^2+4}dx =$

$= 2\ln 8 - 2x\Big|_0^2 + 8\int_0^2 \frac{1\cdot 2}{4(1+(\frac{x}{2})^2)}d\frac{x}{2} = 2\ln 8 - 2(2-0) + 4\operatorname{arctg}\frac{x}{2}\Big|_0^2 =$

$= 2\ln 8 - 4 + 4(\operatorname{arctg} 1 - \operatorname{arctg} 0) = 2\ln 8 - 4 + \frac{4\pi}{4}$

㉚ 2016
$\int_0^{\frac{\pi}{2}} x\cos^2 x\, dx = \int_0^{\frac{\pi}{2}} x\cdot\frac{1+\cos 2x}{2}dx = \int_0^{\frac{\pi}{2}} \frac{x}{2}dx + \frac{1}{2}\int_0^{\frac{\pi}{2}} x\cos 2x\, dx\frac{2}{2} =$

$= \frac{x^2}{2\cdot 2}\Big|_0^{\frac{\pi}{2}} + \frac{1}{2}\cdot\frac{1}{2}\int_0^{\frac{\pi}{2}} x\cos 2x\, d2x = \frac{\pi^2}{16} - 0 + \frac{1}{4}\int_0^{\frac{\pi}{2}} x\, d\sin 2x =$

$= \frac{\pi^2}{16} + \frac{1}{4}x\sin 2x\Big|_0^{\frac{\pi}{2}} - \frac{1}{4}\int_0^{\frac{\pi}{2}} \sin 2x\, dx\frac{2}{2} =$

$= \frac{\pi^2}{16} + \frac{1}{4}\left(\frac{\pi}{2}\underset{0}{\sin\pi} - 0\right) - \frac{1}{4}\left(\frac{-\cos x}{2}\right)\Big|_0^{\frac{\pi}{2}} =$

$= \frac{\pi^2}{16} - \frac{1}{8}\left(\frac{-\cos\pi}{2} + \frac{\cos 0}{2}\right) = \qquad \frac{\pi^2}{16} - \frac{1}{4}$

㉛ 2016
$\int_1^2 x\log_2\frac{1}{x}dx = \int_1^2 x\log_2 e\ln\frac{1}{x}dx = \log_2 e\int_1^2 x\ln\frac{1}{x}dx =$

$= \log_2 e\int_1^2 \ln\frac{1}{x}d\frac{x^2}{2} = \log_2 e\left(\frac{x^2}{2}\cdot\ln\frac{1}{x}\Big|_1^2 - \int_1^2 \frac{x^2}{2}d\ln\frac{1}{x}\right) =$

$= \log_2 e\left(2\ln\frac{1}{2} - \frac{1}{2}\overset{0}{\ln 1} - \int_1^2 \frac{x^2}{2}\cdot\underbrace{x}_{\log}\underbrace{\left(-\frac{1}{x^2}\right)}_{\frac{1}{x}}dx\right) =$

$= \log_2 e\left(2\ln\frac{1}{2} + \int_1^2 x\,dx\right) = \log_2 e\left(2\ln\frac{1}{2} + \frac{x^2}{2}\right)\Big|_1^2 =$

$= \log_2 e\left(2\ln\frac{1}{2} + 2 - \frac{1}{2}\right)$

㉜ 2015
$\int_0^{\frac{1}{2}} \arcsin x\, dx \overset{\text{по части}}{=} x\cdot\arcsin x\Big|_0^{\frac{1}{2}} - \int_0^{\frac{1}{2}} x\, d\arcsin x =$

$= \frac{1}{2}\arcsin\frac{1}{2} - 0\arcsin 0 - \int_0^{\frac{1}{2}} \frac{2x}{2}\frac{1}{\sqrt{1-x^2}}dx =$

$= \frac{1}{2}\frac{\pi}{6} + \int_0^{\frac{1}{2}} \frac{1}{2\sqrt{1-x^2}}d(1-x^2) = \frac{\pi}{12} + \frac{1}{2}\int_0^{\frac{1}{2}} (1-x^2)^{-\frac{1}{2}}d(1-x^2) =$

$= \frac{\pi}{12} + \frac{1}{2}\frac{(1-x^2)^{\frac{1}{2}}}{\frac{1}{2}}\Big|_0^{\frac{1}{2}} = \frac{\pi}{12} + \left(1-\frac{1}{4}\right)^{\frac{1}{2}} - 1^{\frac{1}{2}} = \frac{\pi}{12} + \frac{\sqrt{3}}{2} - 1$

㉝ 2021
$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} e^{|x|}(\sin x + \cos x)dx = I$

интеграл, който е симетричен около нулата

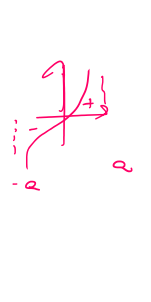
1) разпишем модула

2) разгледаме четността на ф-ията

$\int_{-a}^{a} f(x)dx = \begin{cases} 0, & \text{ако } f(x) \text{ е нечетна} \Leftrightarrow f(-x)=-f(x) \\ 2\int_0^a f(x)dx, & \text{ако } f(x) \text{ е четна} \Leftrightarrow f(-x)=f(x) \end{cases}$

f. нечетна => граф. е симетр. отн. началото на КС

f. четна => граф. е симетр. спрямо Oy



$f(x) = e^{|x|}(\sin x + \cos x)$

$f(-x) = e^{|-x|}(\sin(-x) + \cos(x)) = e^{|x|}(-\sin x + \cos x) \neq \pm f(x)$

=> нито четна, нито нечетна

$g(x) = e^{|x|}\sin x$ - нечетна

$h(x) = e^{|x|}\cos x$ - четна

$\Rightarrow I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} g(x) + h(x)dx = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} g(x)dx + \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} h(x)dx =$

$= 0 + 2\int_0^{\frac{\pi}{2}} e^{|x|}\cos x\, dx = 2\int_0^{\frac{\pi}{2}} e^x\cos x\, dx = \dots$

㉞ 2017
$\int_0^{\frac{\pi}{2}} \left|\frac{1}{2} - \sin^2 x\right| dx \overset{(*)}{=}$

$\left|\frac{1}{2} - \sin^2 x\right| = \begin{cases} \frac{1}{2} - \sin^2 x, & \text{ако } \frac{1}{2} - \sin^2 x \geq 0 \Leftrightarrow \sin^2 x \leq \frac{1}{2} \Rightarrow 0 \leq \sin x \leq \frac{\sqrt{2}}{2} \Rightarrow 0 \leq x \leq \frac{\pi}{4} \\ -\frac{1}{2} + \sin^2 x, & \text{ако } \frac{1}{2} - \sin^2 x < 0 \Leftrightarrow \sin^2 x > \frac{1}{2} \Rightarrow \sin x > \frac{\sqrt{2}}{2} \Rightarrow \frac{\pi}{4} < x < \frac{\pi}{2} \end{cases}$

$0 \leq x \leq \frac{\pi}{2}$, $\sin\frac{\pi}{4}$

$\overset{(*)}{=} \int_0^{\frac{\pi}{4}} \left|\frac{1}{2} - \sin^2 x\right| dx + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \left(\frac{1}{2} - \sin^2 x\right)dx = \int_0^{\frac{\pi}{4}} \frac{1}{2} - \sin^2 x\, dx + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \sin^2 x - \frac{1}{2}dx =$

$= \int_0^{\frac{\pi}{4}} \frac{1}{2} - \frac{1-\cos 2x}{2}dx + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{1-\cos 2x}{2} - \frac{1}{2}dx = \int_0^{\frac{\pi}{4}} \frac{\cos 2x}{2\cdot 2}dx2 + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} -\frac{\cos 2x}{2}\frac{dx}{2} =$

$= \frac{\sin 2x}{4}\Big|_0^{\frac{\pi}{4}} - \frac{\sin 2x}{4}\Big|_{\frac{\pi}{4}}^{\frac{\pi}{2}} = \frac{\sin\frac{\pi}{2}}{4} - \frac{\sin 0}{4} - \left(\frac{\sin\pi}{4} - \frac{\sin\frac{\pi}{2}}{4}\right) =$

$= 2\cdot\frac{1}{4} = \frac{1}{2}$

㉟ 2018
$\int_0^{\frac{\pi}{2}} \frac{dx}{2+\sin x} \overset{(*)}{=}$

универсална субституция:

$t = \operatorname{tg}\frac{x}{2}$

$\operatorname{arctg} t = \frac{x}{2} \Rightarrow x = 2\operatorname{arctg} t$

$\Rightarrow dx = \frac{2}{1+t^2}dt$

$\sin x \overset{\text{ф-ла}}{=} 2\sin\frac{x}{2}\cdot\cos\frac{x}{2} = \frac{2\sin\frac{x}{2}}{\cos\frac{x}{2}}\cdot\cos^2\frac{x}{2} = 2\operatorname{tg}\frac{x}{2}\cdot\cos^2\frac{x}{2} = 2\operatorname{tg}\frac{x}{2}\frac{1}{1+\operatorname{tg}^2\frac{x}{2}} = \frac{2t}{1+t^2}$

$\frac{\cos^2\frac{x}{2}}{\cos^2\frac{x}{2}}$ $\qquad \frac{\sin^2\frac{x}{2}}{\cos^2\frac{x}{2}}$

$\cos x = \cos^2\frac{x}{2} - \sin^2\frac{x}{2} = \frac{\cos^2\frac{x}{2} - \sin^2\frac{x}{2}}{\cos^2\frac{x}{2}}\cdot\cos^2\frac{x}{2} = \left(1 - \operatorname{tg}^2\frac{x}{2}\right)\cos^2\frac{x}{2} = \left(\text{както горе}\right) = \frac{1-t^2}{1+t^2}$

$\overset{(*)}{=} \int_0^1 \frac{1}{2+\frac{2t}{1+t^2}}\frac{2}{1+t^2}dt = \int_0^1 \frac{1+t^2}{1+t^2+t}\cdot\frac{1}{1+t^2}dt = \int_0^1 \overset{\text{рац. ф-ия}}{\frac{1}{t^2+t+1}}dt \overset{\text{субст. на Хорнер}}{=}$

$\boxed{x=0,\ t=\operatorname{tg}\frac{0}{2}=0 \\ x=\frac{\pi}{2},\ t=\operatorname{tg}\frac{\pi}{4}=1}$ $\qquad = \int_{\frac{1}{2}}^{\frac{3}{2}} \frac{1}{y^2+\frac{3}{4}}dy = \int_{\frac{1}{2}}^{\frac{3}{2}} \frac{1}{\frac{3}{4}(\frac{4y^2}{3}+1)}dy =$

$\boxed{t^2+t+1 = t^2+2t\cdot\frac{1}{2}+(\frac{1}{2})^2+\frac{3}{4} = \\ = (t+\frac{1}{2})^2+\frac{3}{4} \\ y=t+\frac{1}{2} \\ dy=dt \\ t=0, y=\frac{1}{2} \\ t=1, y=\frac{3}{2}}$

$= \frac{4}{3}\int_{\frac{1}{2}}^{\frac{3}{2}} \frac{\frac{\sqrt{3}}{2}\cdot 1}{\left((\frac{2}{\sqrt{3}}y)^2+1\right)}d\frac{2}{\sqrt{3}}y =$

$= \frac{4}{3}\cdot\frac{\sqrt{3}}{2}\int_{\frac{1}{2}}^{\frac{3}{2}} \frac{1}{(\frac{2}{\sqrt{3}}y)^2+1}d\left(\frac{2}{\sqrt{3}}y\right) =$

$= \frac{2}{\sqrt{3}}\operatorname{arctg}\left(\frac{2}{\sqrt{3}}y\right)\Big|_{\frac{1}{2}}^{\frac{3}{2}} = \frac{2}{\sqrt{3}}\left(\operatorname{arctg}\sqrt{3} - \operatorname{arctg}\frac{1}{\sqrt{3}}\right) =$

$= \frac{2}{\sqrt{3}}\left(\frac{\pi}{3} - \frac{\pi}{6}\right) = \frac{2}{\sqrt{3}}\cdot\frac{\pi}{6} = \frac{\pi}{3\sqrt{3}}$